DETSI Report

EP Consulting and Dumelle Consulting

Table of contents

Background **1**

Moreton Bay **6**
- The Data . . . 6
- Water Quality and Geographic Modeling . . . 6
  - Model Fit . . . 9
  - Block Kriging . . . 10
  - Alternative Approaches . . . 10
- Resilience Modeling . . . 10
  - Model Fit . . . 10
  - Alternative Approaches . . . 10

Wide Bay **10**
- Outstanding Questions . . . 10
- References . . . 11

Appendix **19**
- Moreton Bay . . . 19
- Wide Bay . . . 19

Background

Accurately characterizing drivers and resilience of seagrass cover is essential to improving the ecological health of the Moreton Bay and Wide Bay ecosystems. Over the past few decades, various scientific efforts have informed our current understanding of trends, behaviors, and flood responses of seagrass cover in both Moreton Bay and Wide Bay [Preen et al., 1995, McMahon et al., 2005, York et al., 2022] [Abal and Dennison, 1996, Udy and Dennison, 1997, Samper-Villarreal et al., 2016, Ovsyanikova et al., 2026] as well as Australia more widely

[Kilminster et al., 2015, Gilby et al., 2018, O'Brien et al., 2018]. Remote sensing has offered new capabilities for seagrass cover mapping [Roelfsema and Phinn, 2009], while innovative statistical approaches have offered novel methods for studying resilience at site-specific scales [Gilby et al., 2023]. Our work contributes to this body of literature by applying spatially explicit statistical models [Cressie, 1993] to study seagrass cover drivers, resilience, and recovery in both Moreton Bay and Wide Bay. Specifically, we provide and contextualize answers to four specific research questions, applicable to both Moreton Bay and Wide Bay:

1. Can we identify water quality covariates that act as triggers for seagrass cover management responses? Are these triggers different based on whether they result from a specific event (e.g., flood) or are ambient?
2. Did seagrass cover improve after the flood event(s)? If so, how and where?
3. Can we identify site-specific resilience indicators, which describe where recovery is likely to be enhanced or impeded?
4. What factors contribute to seagrass resilience?

The spatially explicit statistical models (i.e., spatial models) we used to answer these research questions are related to commonly used linear and generalized linear models but incorporate *spatial dependence* Spatial dependence measures how similar two sites are based on their distance. Formally, we can build spatial dependence directly into spatial models by leveraging Tobler's First Law of Geography [Tobler, 1970], which states succinctly that "nearby sites tend to be more similar than distant sites". Not only is this intuitive based on "first principles" perspective, but spatial models to provide many advantages for modeling spatial data compared to models that assume independence among observations (i.e., nonspatial models). First, nonspatial models tend to underestimate uncertainty, yielding covariate standard errors and p-values for covariates that are too small and corresponding confidence intervals that are too narrow. Second, spatial models improve prediction at unsampled locations by borrowing strength from nearby observations, yielding more precise predictions and corresponding standard errors that vary by location. Third, spatial models decompose model error into separate spatial and nonspatial components, informing residual and resilience structure in ways that nonspatial models cannot. Fourth, spatial models can provide ecologically meaningful insight into the spatial dependence structure of the process, informing relationships among measured and unmeasured variables as well as future sampling plans. Zimmerman and Ver Hoef [2024] provide a thorough review of spatial models for ecological and environmental data, providing more detail and justification for the claims made above.

Spatial models offer many benefits over commonly used machine learning algorithms like random forests [Breiman, 2001, Cutler et al., 2007, James et al., 2017, Kuhn and Silge, 2022]. First, traditional machine learning algorithms do not incorporate spatial dependence and, by consequence, often perform worse than spatial models at characterizing important covariates and predicting at unobserved locations [Fox et al., 2020]. Second, machine learning algorithms struggle to quantify uncertainty in a principled manner, something spatial models excel at. Third, covariate inference is more straightforward using spatial models, as they naturally provide p-values that enable structured testing of hypotheses. Fourth, spatial models provide a

rich set of diagnostics like leverage and Cook's distances [Montgomery et al., 2021], quantities not generally defined for machine learning algorithms. This is not to say that machine learning algorithms cannot be very useful in ecological data analyses, but rather than spatial models are often a more effective tool for modeling spatial data.

Though there are many benefits of spatial models over machine learning algorithms, the reverse is also true. Machine learning algorithms much more simply capture complex nonlinearities "out-of-the-box", bypassing the tuning often required for spatial models that involve nonlinear components like splines. This can be especially useful for ecological data, which are often full of nonlinear relationships [Fox et al., 2017]. Machine learning algorithms also typically rely on few, if any, distributional assumptions and can be more robust against unusual observations like outliers [Hastie et al., 2009]. Machine learning algorithms also tend to be much mroe computational efficient than spatial models. Importantly, machine learning algorithms and spatial models can be complementary tools that help shed insight into complicated ecological processes (something we reiterate in the upcoming analyses). An exciting and developing area of statistical research involves incorporating spatial modeling strategies directly into machine learning algorithms [Saha et al., 2023, Heaton et al., 2025].

The first spatially explicit model we define is the spatial linear model. The spatial linear model is a *mixed-effects model* [Pinheiro and Bates, 2000, Bolker, 2015] written as

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + ... + \beta_p x_{p,i} + \tau_i + \epsilon_i, \tag{1}$$

where $i = 1, 2, ..., n$ indexes the n observations. The quantity y_i is the response variable (for observation i), the β parameters are slope parameters controlling the covariates, $x_{j,i}$ (for $j = 1, 2, ..., p$). The random error τ_i is a spatial random error, and the random error ϵ_i is the nonspatial (i.e., independent) random error. The spatial random error is spatially structured, capturing spatial residual effects like environmental gradients we do not have data to represent. The nonspatial random error is not spatially structure and captures nonspatial residual effects like measurement error and microscale variation. The difference between a traditional (i.e., nonspatial) linear model and a spatial linear model is the inclusion of τ_i (in the spatial linear model). The spatial random error implied by τ_i is governed by a *spatial covariance function*. The spatial covariance function allows the model to incorporate spatial dependence, yielding the improved model performance previously described. Figure 1 shows three different spatial covariance functions that describe different ways the spatial covariance decays with distance. Model parameters and standard errors are estimated from data using likelihood-based [Patterson and Thompson, 1971, Harville, 1977, Wolfinger et al., 1994] or semivariogram-based [Cressie, 1985, Curriero and Lele, 1999] estimation methods. Often, the spatial linear model in Equation 1 in is written more compactly using *matrix notation*:

$$\mathbf{y} = \mathbf{X}\beta + \tau + \epsilon, \tag{2}$$

where $\mathbf{X}$ is a matrix that holds the j covariates for each of n observations, and $\mathbf{y}$, β, τ, and τ are vectors that contain each of the n or j quantities from Equation 1. The spatial linear

model can be adjusted to account for nonspatial random effects just as one would add random effects to a nonspatial linear mixed model:

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}\mathbf{u} + \tau + \epsilon, \tag{3}$$

where $\mathbf{Z}$ is a random effect design matrix (dimension $n \times k$) that indexes each of the k random effects in $\mathbf{u}$. For more information about specific forms of various spatial covariance functions, see Schabenberger and Gotway [2017] and Zimmerman and Ver Hoef [2024].

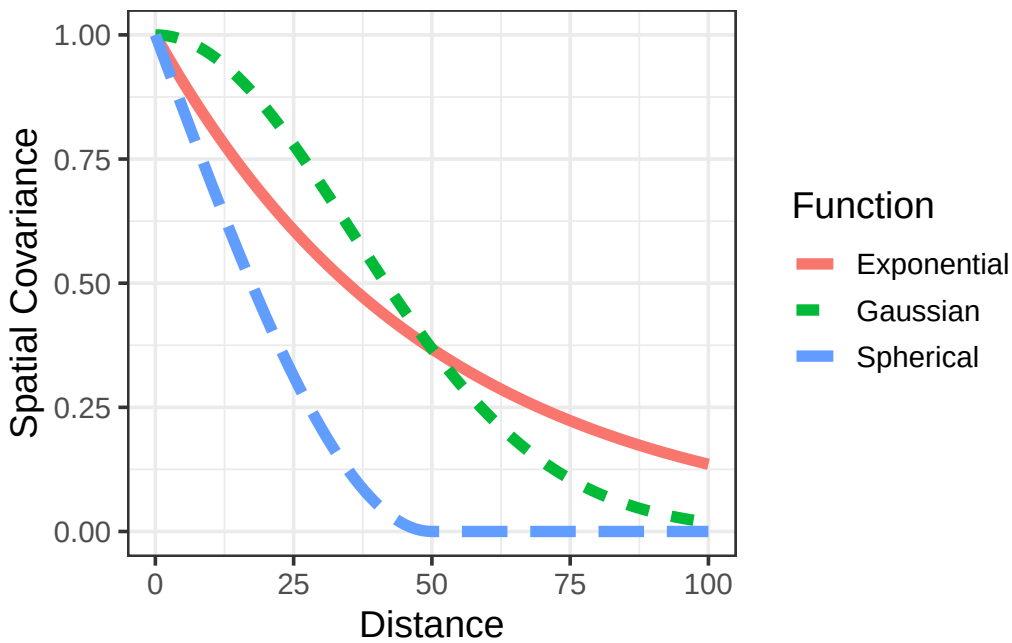

Figure 1: Three spatial covariance functions that represent different spatial dependence behavior.

Like the traditional linear model, the spatial linear model can capture complex nonlinearities in the data through the use of interaction effects [Faraway, 2002, Lenth and Piaskowski, 2025], splines [Chambers and Hastie, 1992], additive models [Wood, 2017, Pedersen et al., 2019], and penalized regression models like the ridge and lasso [Tibshirani, 1996]. Generally, this is accomplished by leveraging *basis functions* that break down nonlinear behavior into a set of additive linear components that may include penalty terms which guard against overfitting. Furthermore, there is no assumption that y_i itself is normally distributed; rather, distributional assumptions are placed on the random errors after incorporating the slope parameters. This clarification implies the spatial (and nonspatial) linear models can be quite effective at modeling a range of data that are not normally distributed. For highly nonnormal data, tools like generalized linear models, which we discuss later, are helpful. Finally, under general conditions, the slope parameters of the spatial (and nonspatial) linear models are approximately

normally distributed with a large sample size (see Casella and Berger [2024] for a general justification via the Central Limit Theorem and Zhang and Zimmerman [2005] for nuances regarding spatial data). This is beneficial because it justifies valid hypothesis testing on the slope coefficients (often of primary ecological interest), no matter the distribution of the data (given certain assumptions are satisfied).

One of the most useful applications of the spatial linear model is predicting the response variable at unobserved locations. A very useful spatial prediction method called Kriging has a rich theoretical justification [Cressie, 1990, Stein, 1999]. Kriging is the spatial analogue to best linear unbiased prediction (BLUP) for mixed models [Henderson, 1975]. Via Kriging, the spatial linear model produces BLUPs at each unobserved location of interest, alongside prediction intervals to characterize uncertainty. While useful, sometimes the goal is to create a BLUP for an entire region of interest. This technique, which builds upon the theoretical foundations of Kriging, is called block Kriging [Ver Hoef, 2002, Cressie, 2006]. The intuition is to make point predictions at a very dense grid in the study region and then average these predictions, appropriately characterizing uncertainty. Together, Kriging and block Kriging provide a complete set of tools for spatial prediction, given the fit of a spatial linear model.

Spatial generalized linear models are an extension of spatial linear models to highly nonnormal data [Christensen and Waagepetersen, 2002, Zhang, 2002, Diggle and Ribeiro Jr, 2007, Bonat and Ribeiro Jr, 2016]. Ver Hoef et al. [2024] proposed a novel application of the Laplace approximation to enable restricted maximum likelihood estimation of spatial generalized linear models. Their model formulation builds upon Equation 1 and is given by

$$g(\mu_i) = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + ... + \beta_p x_{p,i} + \tau_i + \epsilon_i, \tag{4}$$

where $g(\mu_i)$ is a "link function" that links the distribution of y_i to a linear function of its mean, μ_i. Common link functions include the log link for count and skewed data and the logit link for binomial and proportion data (Table 1). Using the same argument as for spatial linear models, nonspatial random effects can be added to spatial generalized linear models.

Table 1: Common generalized linear model families, link functions, and appropriate data types.

Family	Link Function	Link Name	Data Type
Binomial	$f(\mu) = \log(\mu/(1-\mu))$	Logit	Binary; Binary Count
Poisson	$f(\mu) = \log(\mu)$	Log	Count
Negative Binomial	$f(\mu) = \log(\mu)$	Log	Count
Beta	$f(\mu) = \log(\mu/(1-\mu))$	Logit	Proportion
Gamma	$f(\mu) = \log(\mu)$	Log	Skewed
Inverse Gaussian	$f(\mu) = \log(\mu)$	Log	Skewed

Spatial models are complicated, both theoretically and computationally. This has contributed to their relative lack of use in the ecological community. However, recent advances in open-

source software have made spatial models more accessible to ecologists who have strong subject matter expertise but may lack sufficient statistical background or computational skills to implement them from scratch. One such advancement is the `spmodel` **R** package [Dumelle et al., 2023], which emulates base-**R** functions like `lm()` and `glm()` to account for spatial dependence via `splm()` and `spglm()`, respectively. Familiar **R** functions like `summary()` and `predict()` acn be used directly on models fit via `spmodel`. Importantly, `spmodel` also has tools for spatial random forest residual modeling [Fox et al., 2020], applications to large data of several thousand observations via spatial indexing [Ver Hoef et al., 2023], and cross validation [Stone, 1974, Roberts et al., 2017].

Moreton Bay

The Data

We started with just over 10,000 seagrass cover observations in Moreton Bay across three financial years (2020-2021, 2022-2023, and 2023-2024). eReefs data [Steven et al., 2019] with modeled water quality covariates related to turbidity and salinity were only available at 7,657 observaations. Because one of our research questions involved connecting water quality covariates to management actions, we proceed with this subset of 7,657 observations for modeling. Of the 7,657 observations, 1,995 were in 2020-2021 (pre-flood), 3,329 were in 2022-2023, and 2,333 were in 2023-2024 (Table 2).

Table 2

Financial Year	Sample Size	Percent (of Total)
2020-2021	1,995	26.1%
2022-2023	3,329	43.5%
2023-2024	2,333	30.5%
Total	7,657	100.0%

Total seagrass observations Moreton Bay by financial year.

Water Quality and Geographic Modeling

Total cover is a percentage that ranges from 0 (no seagrass) to 100 (seagrass entirely contained in an area). Figure 2 shows total cover by financial year throughout the Bay. Total cover tends to be largest along the eastern shoreline of the Bay and lowest in the middle of the Bay. Figure 2 also reveals that 2022-2023 had by far the most sampling in the middle of the Bay.

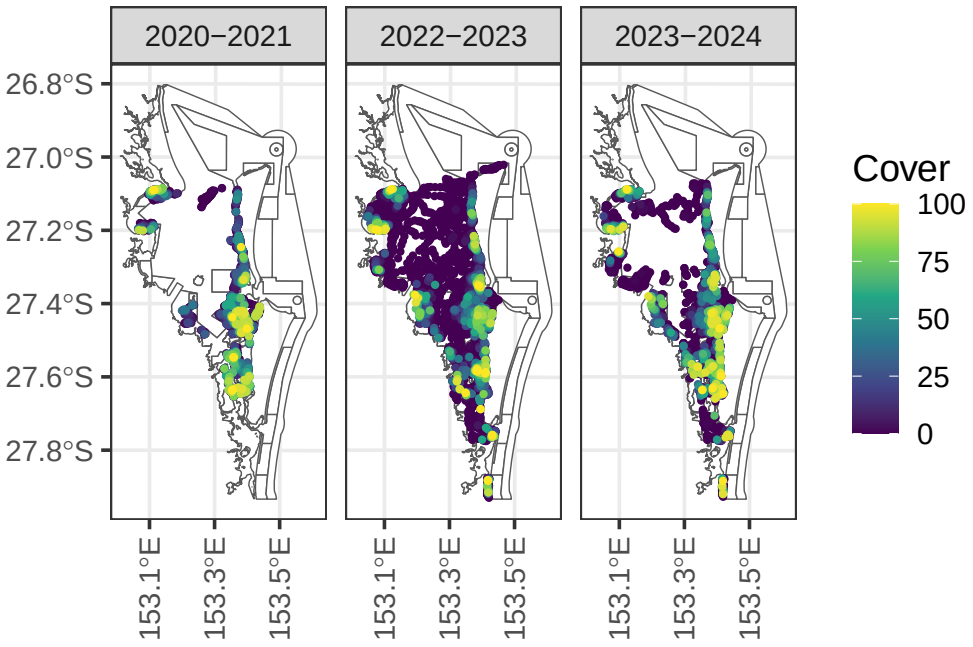

Figure 2: Total seagrass cover in Moreton Bay by financial year.

We considered for modeling several water quality and geographic covariates. Table 3 summarize the water quality variables, which included both ambient and flood-related metrics related to turbidity, salinity, and temperature. The "Variable" column indicates the water quality variable turbidity, salinity, or temperature. The "Flood" variable indicates whether the measurement corresponds to a flood variable or not. The "Period" variable indicates the prior period (in days) relative to sampling or indicates whether the variable corresponds to a flood event. The "Type" column is the type of statistic, either a mean, minimum, maximum, number of days above (or below) a cutoff during "Period", or a maximum spell (i.e., duration) above (or below) a cutoff during "Period". The "Included" variable indicates whether the variable was included as a covariate in the final spatial model. We included only a few variables in the final model: mean turbidity in the 90 days prior to sampling, mean turbidity during the flood event, and mean temperature in the 90 days prior to sampling. We only included a single turbidity metric prior to sampling because these variables were all quite correlated with one another, making it hard for the model to tease apart signals. We believed 90 day mean turbidity prior to sampling to be the most accurate "running average" of turbidity prior to sampling. We used a similar rationale to select only a single turbidity metric from the flood and a single temperature metric prior to sampling. We omitted a flood salinity metric because it did not appear to have a significant impact on the total cover at modeling, and additionally, because we lacked data for salinity prior to sampling, it would be challenging to establish a "baseline" for salinity in order to tease apart prior to sampling effects from flood effects. We provide more detail regarding the modeling in Section .

Table 3: Moreton Bay water quality variables considered for modeling.

Variable	Flood	Period	Type	Cutoff	Included
Turbidity	No	7	Mean	NA	No
Turbidity	No	30	Mean	NA	No
Turbidity	No	90	Mean	NA	Yes
Turbidity	No	7	Above	3	No
Turbidity	No	30	Above	3	No
Turbidity	No	90	Above	3	No
Turbidity	No	7	Above	5	No
Turbidity	No	30	Above	5	No
Turbidity	No	90	Above	5	No
Turbidity	No	7	Above	6	No
Turbidity	No	30	Above	6	No
Turbidity	No	90	Above	6	No
Turbidity	Yes	Flood	Above	3	No
Turbidity	Yes	Flood	Above	5	No
Turbidity	Yes	Flood	Above	6	No
Turbidity	Yes	Flood	Duration	3	No
Turbidity	Yes	Flood	Duration	5	No
Turbidity	Yes	Flood	Duration	6	No
Turbidity	Yes	Flood	Mean	NA	Yes
Turbidity	Yes	Flood	Max	NA	No
Salinity	No	7	Below	20	No
Salinity	No	30	Below	20	No
Salinity	No	90	Below	20	No
Salinity	No	7	Below	25	No
Salinity	No	30	Below	25	No
Salinity	No	90	Below	25	No
Salinity	No	7	Below	30	No
Salinity	No	30	Below	30	No
Salinity	No	90	Below	30	No
Salinity	Yes	Flood	Below	20	No
Salinity	Yes	Flood	Below	25	No
Salinity	Yes	Flood	Below	30	No
Salinity	Yes	Flood	Duration	20	No
Salinity	Yes	Flood	Duration	25	No
Salinity	Yes	Flood	Duration	30	No
Salinity	Yes	Flood	Mean	NA	No
Salinity	Yes	Flood	Min	NA	No
Temperature	No	7	Mean	NA	No
Temperature	No	30	Mean	NA	No

Table 3: Moreton Bay water quality variables considered for modeling.

Variable	Flood	Period	Type	Cutoff	Included
Temperature	No	90	Mean	NA	Yes

Table 4 summarizes the geographic variables, which have the following structure:

- The "Sediment Type" variable has six levels: Sand, Sandy Mud, Muddy Sand, Mud, Rock, and Other.
- The "Tidal Range" variable has three levels: Shallow Subtidal, Intertidal, and Deep Subtidal.
- The "Years Since Flood" is numeric from zero (pre flood) to 1.93 years post flood (February, 2024).
- The "Nearest River" variable has six levels: Brisbane River, Caboolture River, Logan River, Ningi Creek, Pimpama Coomera and Nerang Rivers, and Pine River.
- The "Distance to (Nearest River) Outlet (in km)" variable is numeric from 0.24 to 34.97.
- The "Growing Season" variable has two levels: Growing (September - December) and Not Growing (January - August).
- The EHMP subzone variable has eleven levels: Bramble Bay, Broadwater, Central Bay, Deception Bay, Eastern Banks, Moreton Bay Open Coastal Waters, North-Eastern Moreton Bay, Northern Banks Open Coastal Waters, South-Eastern Moreton Bay, Southern Bay, and Waterloo Bay (Figure 3).

Table 4

Variable	Effect	Included
Sediment Type	Fixed	Yes
Tidal Range	Fixed	Yes
Years Since Flood	Fixed	Yes
Nearest River	Fixed	No
Distance to Outlet	Fixed	No
Growing Season	Fixed	No
EHMP Subzone	Random	Yes

Moreton Bay geographic variables considered for modeling.

From Table 4, we kept the sediment type, tidal range, and years since flood variables as covariates in the model. The nearest river and distance to nearest outlet variables were removed, as they did not seem related to total cover. We removed growing season from the model after determining that it likely it not ecologically useful, as a simple month proxy likely does not adequately account for the complexities of the more nuanced aspects of seagrass growth with

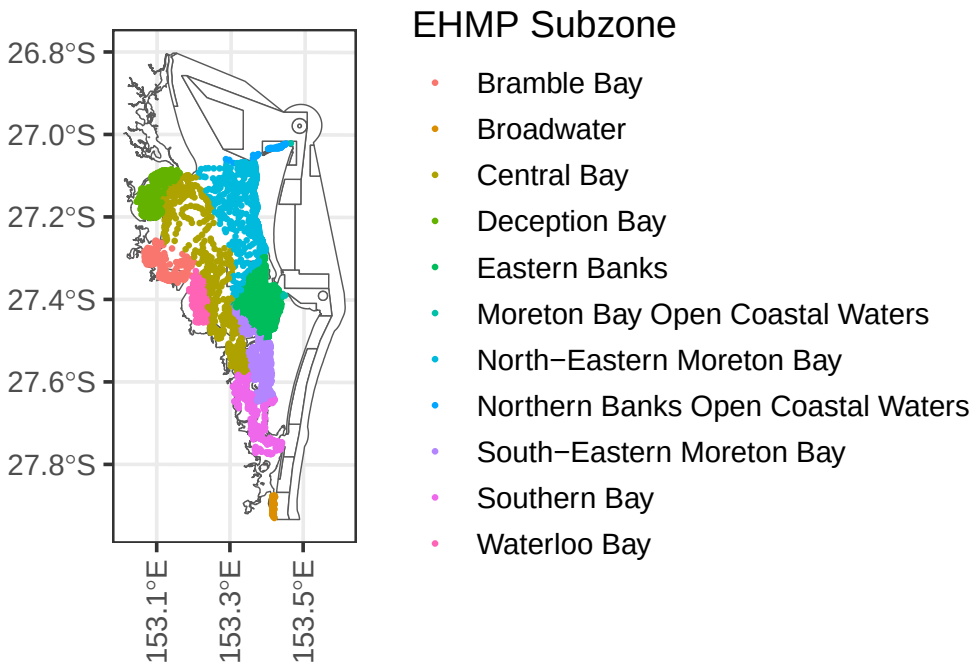

Figure 3: EHMP Subzones in Moreton Bay.

time. We used EHMP subzone as a random effect to account for within-subzone correlation that may impact the relationships among total cover and the remaining variables. We provide more detail regarding the modeling in Section .

Model Fit

Table 5: ANOVA table of covariates in the Moreton Bay water quality and geographic model.

effects	df	statistic	p.value
depth.range	2	613.833	0.000
sed.type	5	109.286	0.000
temp.90d.mn	1	45.708	0.000
yr.flood	1	31.765	0.000
turb.fld.mn:yr.flood.recip2	1	28.968	0.000
after.flood	1	28.950	0.000
(Intercept)	1	28.470	0.000
turb.90d.mn	1	4.332	0.037

Table 6: Variance components in the Moreton Bay water quality and geographic model.

Variance Component	Percent (of Total Variability)
Covariates	11.6%
Spatial Effect	31.5%
Independent Error	49.7%
EHMP Subzone	7.3%

Block Kriging

Alternative Approaches

Resilience Modeling

Model Fit

Alternative Approaches

Wide Bay

Outstanding Questions

- What are the units for the turbidity/salinity/temperature variables?

References

E. G. Abal and W. C. Dennison. Seagrass depth range and water quality in southern Moreton Bay, Queensland, Australia. *Marine and Freshwater Research*, 47(6):763–771, 1996. doi: 10.1071/MF9960763.

Benjamin M. Bolker. Linear and generalized linear mixed models. In Gordon A. Fox, Simoneta Negrete-Yankelevich, and Vinicio J. Sosa, editors, *Ecological Statistics: Contemporary Theory and Application*, pages 309–333. Oxford University Press, Oxford, 2015. doi: 10.1093/acprof:oso/9780199672547.003.0014.

Wagner Hugo Bonat and Paulo Justiniano Ribeiro Jr. Practical likelihood analysis for spatial generalized linear mixed models. *Environmetrics*, 27(2):83–89, 2016. doi: 10.1002/env.2375.

Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001. doi: 10.1023/A:1010933404324.

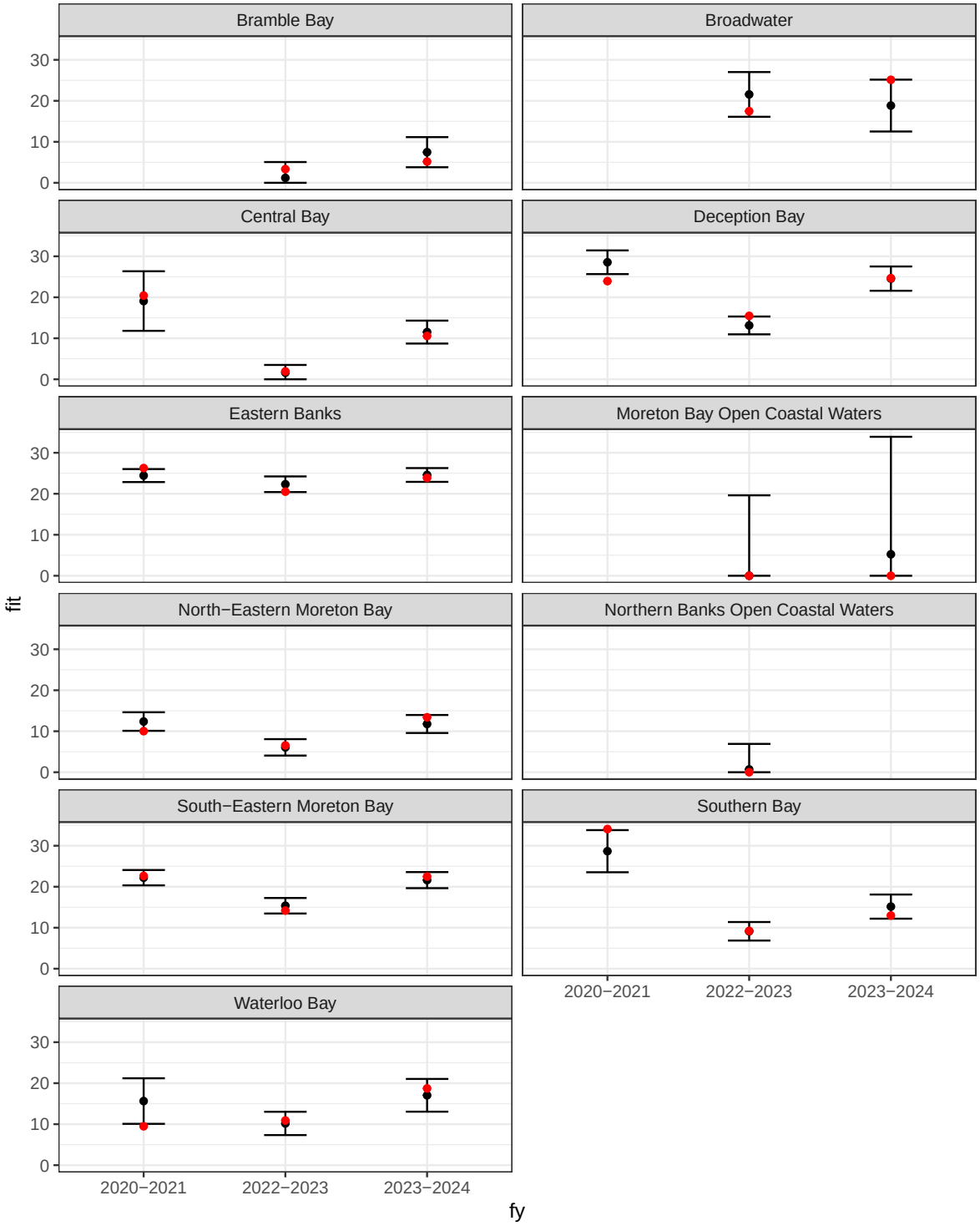

Figure 4: Average total cover predictions by EHMP subzone. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

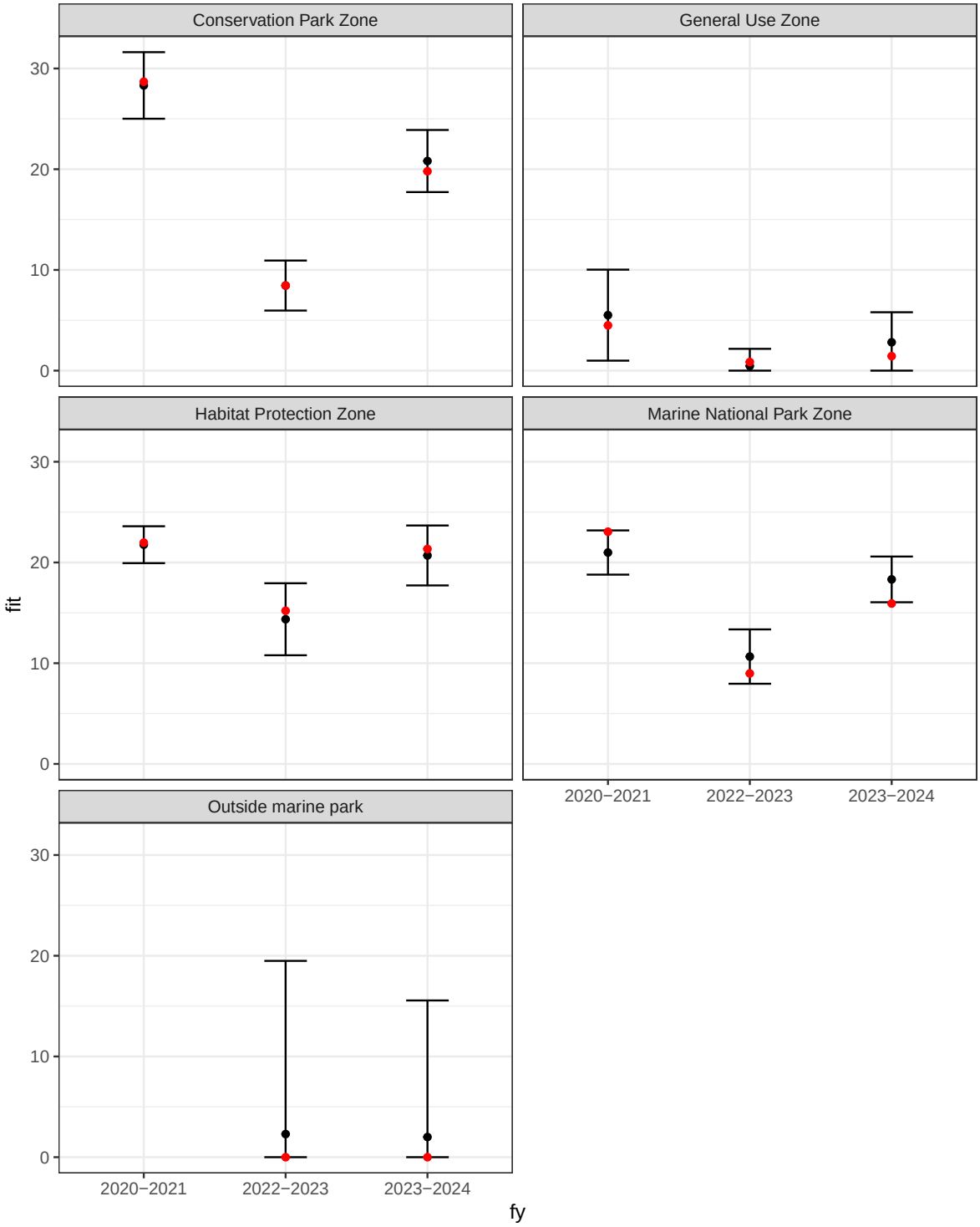

Figure 5: Average total cover predictions by EHMP convervation zone. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

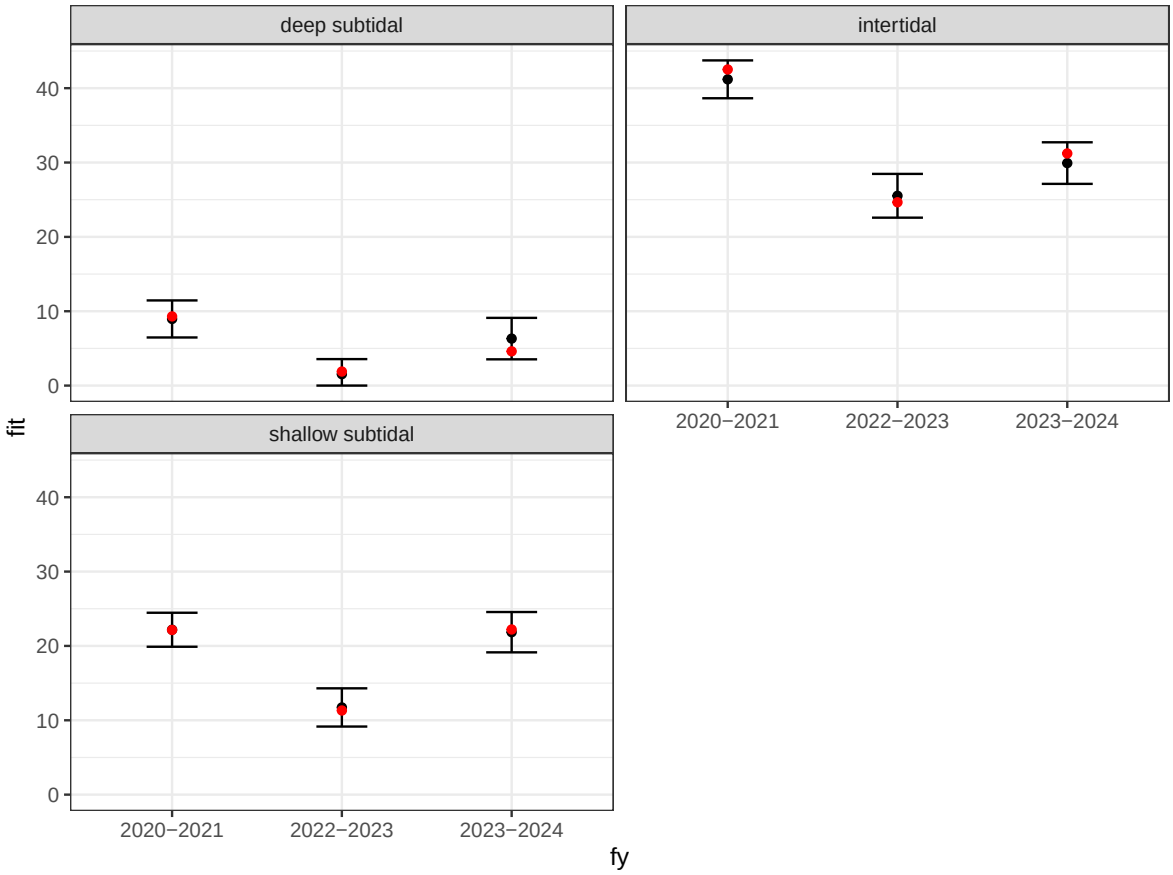

Figure 6: Average total cover predictions by EHMP tidal range. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

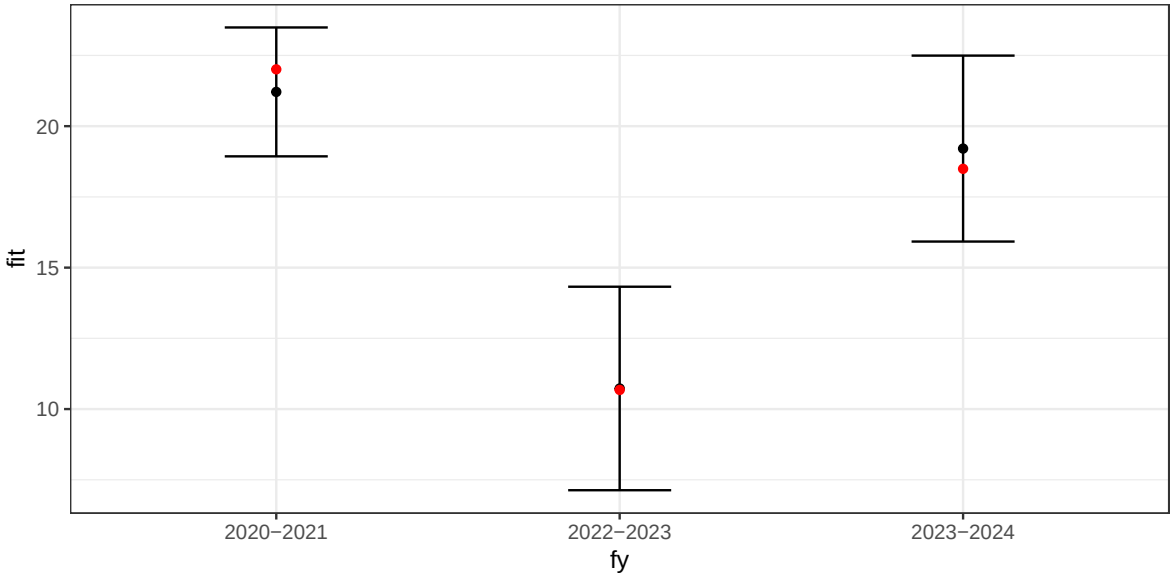

Figure 7: Average total cover predictions by Moreton Bay. Red dots indicate the mean from the raw data. Black dots indicate the predicted mean in the region alongside a prediction interval.

George Casella and Roger L. Berger. *Statistical Inference*. Chapman and Hall/CRC, 2nd edition, 2024.

John M. Chambers and Trevor J. Hastie. *Statistical Models in S*. Routledge, 1992.

Ole F. Christensen and Rasmus Waagepetersen. Bayesian prediction of spatial count data using generalized linear mixed models. *Biometrics*, 58(2):280–286, 2002. doi: 10.1111/j.0006-341X.2002.00280.x.

Noel Cressie. Fitting variogram models by weighted least squares. *Journal of the International Association for Mathematical Geology*, 17(5):563–586, 1985. doi: 10.1007/BF01032109.

Noel Cressie. The origins of kriging. *Mathematical Geology*, 22(3):239–252, 1990. doi: 10.1007/BF00889887.

Noel Cressie. *Statistics for Spatial Data*. Wiley, Hoboken, NJ, 1993. doi: 10.1002/9781119115151.

Noel Cressie. Block kriging for lognormal spatial processes. *Mathematical Geology*, 38(4):413–443, 2006. doi: 10.1007/s11004-005-9022-8.

Frank C. Curriero and Subhash Lele. A composite likelihood approach to semivariogram estimation. *Journal of Agricultural, Biological, and Environmental Statistics*, 4(1):9–28, 1999.

D. Richard Cutler, Thomas C. Edwards, Karen H. Beard, Adele Cutler, Kyle T. Hess, Jacob Gibson, and Joshua J. Lawler. Random forests for classification in ecology. *Ecology*, 88(11): 2783–2792, 2007. doi: 10.1890/07-0539.1.

Peter J. Diggle and Paulo Justiniano Ribeiro Jr. *Model-Based Geostatistics*. Springer Series in Statistics. Springer, New York, 2007. ISBN 9780387329072. doi: 10.1007/978-0-387-48536-2.

Michael Dumelle, Matt Higham, and Jay M. Ver Hoef. spmodel: Spatial statistical modeling and prediction in R. *PLOS ONE*, 18(3):e0282524, 2023. doi: 10.1371/journal.pone.0282524.

Julian James Faraway. *Practical Regression and ANOVA Using R*. University of Bath, Bath, 2002. URL https://cran.r-project.org/doc/contrib/Faraway-PRA.pdf.

Eric W. Fox, Ryan A. Hill, Scott G. Leibowitz, Anthony R. Olsen, Darren J. Thornbrugh, and Marc H. Weber. Assessing the accuracy and stability of variable selection methods for random forest modeling in ecology. *Environmental Monitoring and Assessment*, 189(7):316, 2017. doi: 10.1007/s10661-017-6025-0.

Eric W. Fox, Jay M. Ver Hoef, and Anthony R. Olsen. Comparing spatial regression to random forests for large environmental data sets. *PLOS ONE*, 15(3):e0229509, 2020. doi: 10.1371/journal.pone.0229509.

Ben L. Gilby, Andrew D. Olds, Rod M. Connolly, Paul S. Maxwell, Christopher J. Henderson, and Thomas A. Schlacher. Seagrass meadows shape fish assemblages across estuarine seascapes. *Marine Ecology Progress Series*, 588:179–189, 2018. doi: 10.3354/meps12394.

Ben L. Gilby, Lucy A. Goodridge Gaines, Hayden P. Borland, Christopher J. Henderson, Jesse D. Mosman, Andrew D. Olds, and Hannah J. Perry. Drivers of ecological condition identify bright spots and sites for management across coastal seascapes. *Estuaries and Coasts*, 46(4):906–924, 2023. doi: 10.1007/s12237-022-01164-z.

David A. Harville. Maximum likelihood approaches to variance component estimation and to related problems. *Journal of the American Statistical Association*, 72(358):320–338, 1977. doi: 10.1080/01621459.1977.10480998.

Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer Series in Statistics. Springer, 2nd edition, 2009. doi: 10.1007/978-0-387-84858-7. URL https://hastie.su.domains/ElemStatLearn/.

Matthew J. Heaton, Andrew Millane, and Jake S. Rhodes. A scalable spatial decorrelation preprocessing approach for machine and deep learning. *Journal of Data Science*, pages 1–15, 2025.

Charles R. Henderson. Best linear unbiased estimation and prediction under a selection model. *Biometrics*, 31(2):423–447, 1975. doi: 10.2307/2529430.

Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani. *An Introduction to Statistical Learning with Applications in R*. Springer, 2017. doi: 10.1007/978-1-4614-7138-7.

Kieryn Kilminster, Kathryn McMahon, Michelle Waycott, Gary A. Kendrick, Peter Scanes, Len McKenzie, Katherine R. O'Brien, Mitchell Lyons, Angus Ferguson, Paul Maxwell, et al. Unravelling complexity in seagrass systems for management: Australia as a microcosm. *Science of the Total Environment*, 534:97–109, 2015. doi: 10.1016/j.scitotenv.2015.04.061.

Max Kuhn and Julia Silge. *Tidy Modeling with R: A Framework for Modeling in the Tidyverse*. O'Reilly Media, 2022. URL https://www.tmwr.org.

Russell V. Lenth and Julia Piaskowski. *emmeans: Estimated Marginal Means, aka Least-Squares Means*, 2025. URL https://CRAN.R-project.org/package=emmeans. R package version 2.0.1.

Kathryn McMahon, Susan Bengtson Nash, Geoff Eaglesham, Jochen F. Müller, Norman C. Duke, and Steve Winderlich. Herbicide contamination and the potential impact to seagrass meadows in Hervey Bay, Queensland, Australia. *Marine Pollution Bulletin*, 51(1–4):325–334, 2005. doi: 10.1016/j.marpolbul.2004.10.048.

Douglas C. Montgomery, Elizabeth A. Peck, and G. Geoffrey Vining. *Introduction to Linear Regression Analysis*. John Wiley & Sons, 6th edition, 2021.

Katherine R. O'Brien, Michelle Waycott, Paul Maxwell, Gary A. Kendrick, James W. Udy, Angus J. P. Ferguson, Kieryn Kilminster, Peter Scanes, Len J. McKenzie, Kathryn McMahon, et al. Seagrass ecosystem trajectory depends on the relative timescales of resistance, recovery and disturbance. *Marine Pollution Bulletin*, 134:166–176, 2018. doi: 10.1016/j.marpolbul.2017.09.006.

E. Ovsyanikova, W. N. Venables, and J. W. Udy. Spatial and temporal modelling of seagrass distribution in Moreton Bay, Australia, based on long-term citizen science data. *Estuarine, Coastal and Shelf Science*, 328:109610, 2026. doi: 10.1016/j.ecss.2025.109610.

H. Desmond Patterson and Robin Thompson. Recovery of inter-block information when block sizes are unequal. *Biometrika*, 58(3):545–554, 1971. doi: 10.1093/biomet/58.3.545.

Eric J. Pedersen, David L. Miller, Gavin L. Simpson, and Noam Ross. Hierarchical generalized additive models in ecology: an introduction with mgcv. *PeerJ*, 7:e6876, 2019. doi: 10.7717/peerj.6876.

José C. Pinheiro and Douglas M. Bates. *Mixed-Effects Models in S and S-PLUS*. Springer, 2000. doi: 10.1007/b98882.

A. R. Preen, W. J. Lee Long, and R. G. Coles. Flood and cyclone related loss, and partial recovery, of more than 1000 km^2 of seagrass in Hervey Bay, Queensland, Australia. *Aquatic Botany*, 52(1–2):3–17, 1995. doi: 10.1016/0304-3770(95)00491-H.

David R. Roberts, Volker Bahn, Simone Ciuti, Mark S. Boyce, Jane Elith, Gurutzeta Guillera-Arroita, Severin Hauenstein, José J. Lahoz-Monfort, Boris Schröder, Wilfried Thuiller, et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929, 2017. doi: 10.1111/ecog.02881.

C. M. Roelfsema and S. R. Phinn. An integrated field and remote sensing approach for mapping seagrass cover, Moreton Bay, Australia. *Journal of Spatial Science*, 54(1):45–62, 2009. doi: 10.1080/14498596.2009.9635166.

Arkajyoti Saha, Sumanta Basu, and Abhirup Datta. Random forests for spatially dependent data. *Journal of the American Statistical Association*, 118(541):665–683, 2023. doi: 10.1080/01621459.2021.1950003.

Jimena Samper-Villarreal, Catherine E. Lovelock, Megan I. Saunders, Chris Roelfsema, and Peter J. Mumby. Organic carbon in seagrass sediments is influenced by seagrass canopy complexity, turbidity, wave height, and water depth. *Limnology and Oceanography*, 61(3): 938–952, 2016. doi: 10.1002/lno.10262.

Oliver Schabenberger and Carol A. Gotway. *Statistical Methods for Spatial Data Analysis*. Chapman and Hall/CRC, 2017. doi: 10.1201/9781315275086.

Michael L. Stein. *Interpolation of Spatial Data: Some Theory for Kriging*. Springer, 1999. doi: 10.1007/978-1-4612-1494-6.

Andrew D. L. Steven, Mark E. Baird, Richard Brinkman, Nicholas J. Car, Simon J. Cox, Mike Herzfeld, Jonathan Hodge, Emlyn Jones, Edward King, Nugzar Margvelashvili, et al. eReefs: An operational information system for managing the Great Barrier Reef. *Journal of Operational Oceanography*, 12(sup2):S12–S28, 2019. doi: 10.1080/1755876X.2019.1656720.

Mervyn Stone. Cross-validatory choice and assessment of statistical predictions. *Journal of the Royal Statistical Society: Series B (Methodological)*, 36(2):111–133, 1974. doi: 10.1111/j.2517-6161.1974.tb00994.x.

Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 58(1):267–288, 1996. doi: 10.1111/j.2517-6161.1996.tb02080.x.

Waldo R. Tobler. A computer movie simulating urban growth in the Detroit region. *Economic Geography*, 46(sup1):234–240, 1970. doi: 10.2307/143141.

James W. Udy and William C. Dennison. Growth and physiological responses of three seagrass species to elevated sediment nutrients in Moreton Bay, Australia. *Journal of Experimental Marine Biology and Ecology*, 217(2):253–277, 1997. doi: 10.1016/S0022-0981(97)00060-3.

Jay Ver Hoef. Sampling and geostatistics for spatial data. *Ecoscience*, 9(2):152–161, 2002. doi: 10.1080/11956860.2002.11682701.

Jay M. Ver Hoef, Michael Dumelle, Matt Higham, Erin E. Peterson, and Daniel J. Isaak. Indexing and partitioning the spatial linear model for large data sets. *PLOS ONE*, 18(11): e0291906, 2023. doi: 10.1371/journal.pone.0291906.

Jay M. Ver Hoef, Eryn Blagg, Michael Dumelle, Philip M. Dixon, Dale L. Zimmerman, and Paul Conn. Marginal inference for hierarchical generalized linear mixed models with patterned covariance matrices using the Laplace approximation. *Environmetrics*, 35(7):e2872, 2024. doi: 10.1002/env.2872.

Russ Wolfinger, Randy Tobias, and John Sall. Computing Gaussian likelihoods and their derivatives for general linear mixed models. *SIAM Journal on Scientific Computing*, 15(6): 1294–1310, 1994. doi: 10.1137/0915079.

Simon N. Wood. *Generalized Additive Models: An Introduction with R*. Chapman and Hall/CRC, 2nd edition, 2017. doi: 10.1201/9781315370279.

Paul H. York, Catherine V. Bryant, and Michael Rasheed. *Post-Flood Seagrass Monitoring in Hervey Bay May 2022*. Department of Environment and Science, 2022.

Hao Zhang. On estimation and prediction for spatial generalized linear mixed models. *Biometrics*, 58(1):129–136, 2002. doi: 10.1111/j.0006-341X.2002.00129.x.

Hao Zhang and Dale L. Zimmerman. Towards reconciling two asymptotic frameworks in spatial statistics. *Biometrika*, 92(4):921–936, 2005. doi: 10.1093/biomet/92.4.921.

Dale L. Zimmerman and Jay M. Ver Hoef. *Spatial Linear Models for Environmental Data*. Chapman and Hall/CRC, 2024. doi: 10.1201/9781003044680.

Appendix

Moreton Bay

Wide Bay